

Bridging the Semantic Gap: Enhancing Neural Binary-to-IR Lifting with Retrieval-Augmented Generation

Supplementary Material

APPENDIX A: IMPACT OF INPUT LENGTH ON DECOMPILE PERFORMANCE

Figure 1 presents BRIDGE’s decompilation performance across optimization levels (-O0 to -O3) and input assembly code lengths (0-400 instructions) on the HumanEval-Decompile dataset. We observe that re-executability drops significantly from -O1 to -O3 for longer programs, indicating that aggressive optimizations (e.g., inlining, loop transformations) introduce functional errors that BRIDGE struggles to resolve. Edit similarity (average 51.4% across optimizations) is strong, particularly at -O0 (63.6%), but declines between 140-395 instructions (e.g., 31.55% at -O1). This suggests BRIDGE effectively captures high-level semantics for moderately complex binaries but faces challenges with very large, optimized code. Furthermore, CFG similarity is stable across -O0 to -O3, with -O0 slightly higher, suggesting BRIDGE’s structure-aware RAG database effectively preserves control flow despite optimizations.

APPENDIX B: DECOMPILE DEMO: RAG-AUGMENTED AND STATIC ERROR-GUIDED REFINEMENT

Figure 2 compares decompilation performance from assembly to LLVM IR using DeepSeek-V3, DeepSeek-V3 with our proposed structure-aware RAG scheme, BRIDGE (1-pass and 3-pass), and the ground truth, highlighting the impact of our proposed Structure-aware RAG and feedback-directed iterative inferences. Lines in red indicate errors, while lines in blue show the modifications made by the LLM. The ground truth LLVM IR computes the sum of integers from 0 to 99. DeepSeek-V3 suffers from critical structural errors and low edit similarity (29.1%), including the use of undefined values (%t1) and multiple definitions (%edi). Incorporating our structure-aware RAG scheme improves structural coherence, increasing edit similarity to 40.1%, although SSA violations persist. BRIDGE achieves the most substantial improvements through static error analysis and multi-pass refinement, generating recompilable code with edit similarity increasing to 50.4% (1-pass with feedback) and 58.2% (3-pass with feedback). CFG similarity remains consistently above 93%, demonstrating that combining retrieval-based context with feedback-driven correction significantly enhances the structural and functional quality of decompiled LLVM IR.

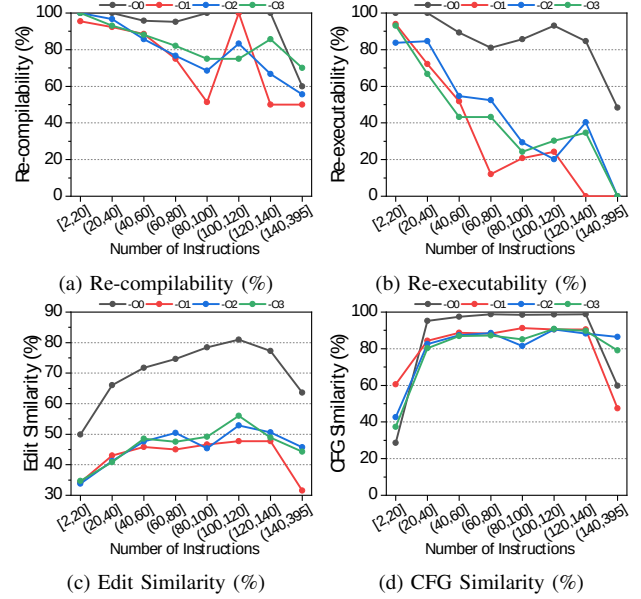


Figure 1: BRIDGE’s performance across varying input assembly code lengths and optimization levels.

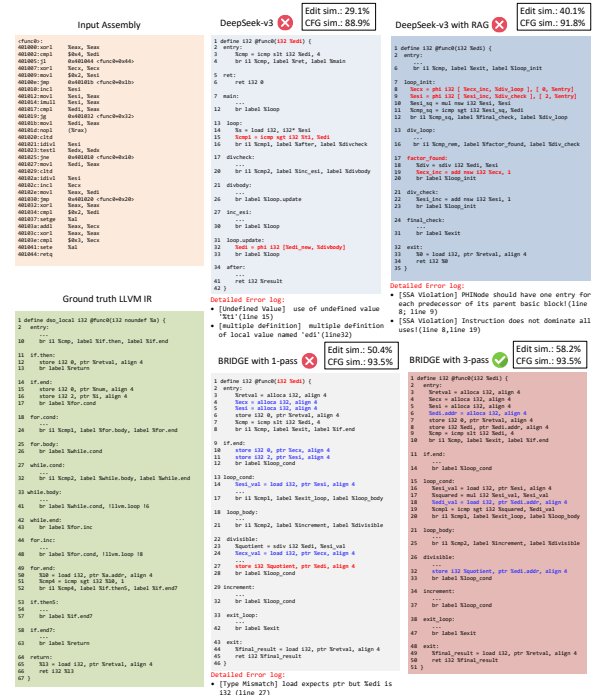


Figure 2: A decompilation demo